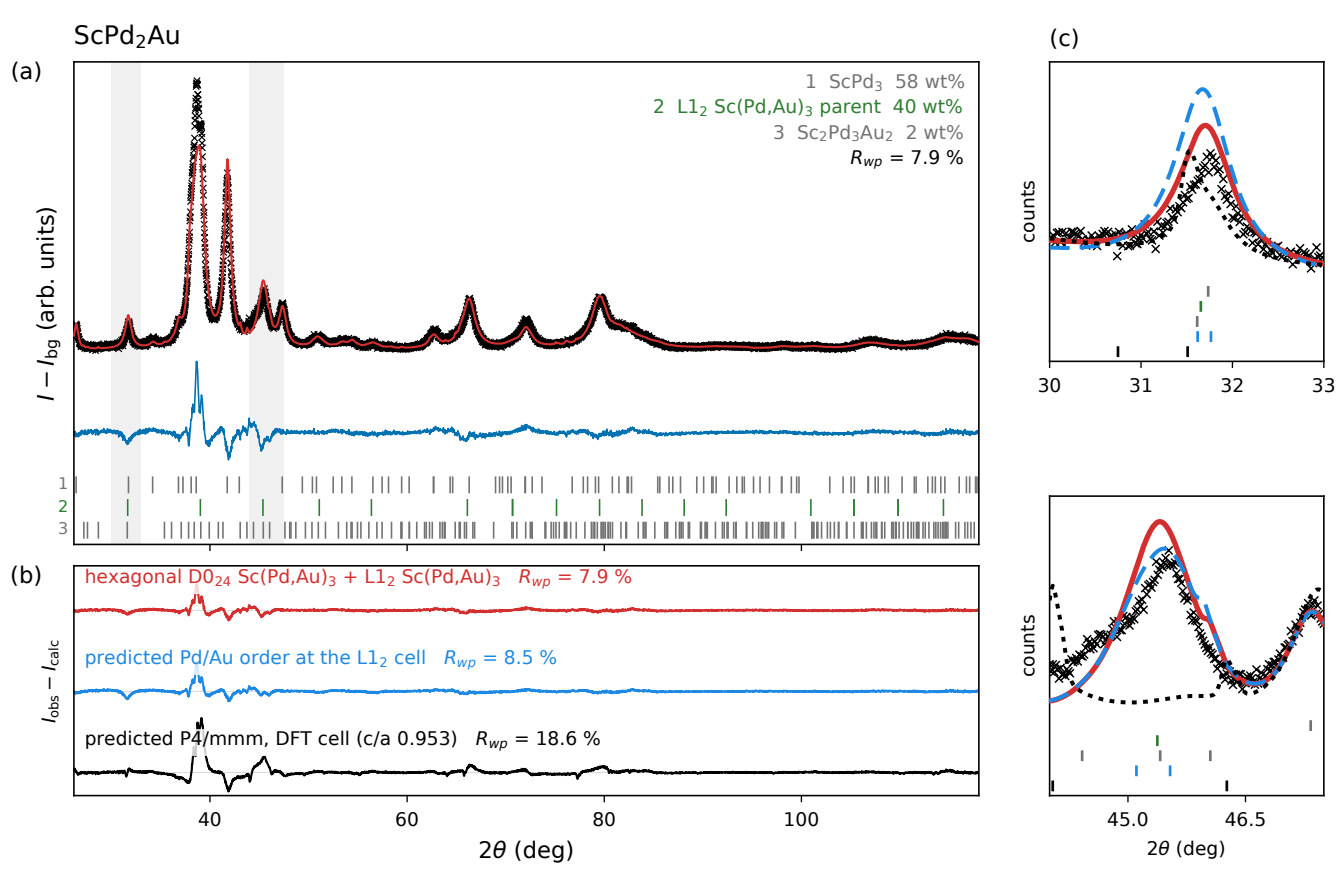

ScPd2Au
(a)
1 ScPd3 58 wt%
2 L12 Sc(Pd,Au)3 parent 40 wt%
3 Sc2Pd3Au2 2 wt%
Rwp = 7.9 %
I − Ibg (arb. units)
1
2
3
(b)
hexagonal D024 Sc(Pd,Au)3 + L12 Sc(Pd,Au)3 Rwp = 7.9 %
predicted Pd/Au order at the L12 cell Rwp = 8.5 %
predicted P4/mmm, DFT cell (c/a 0.953) Rwp = 18.6 %
Iobs − Icalc
40
60
80
100
2θ (deg)
(c)
counts
30
31
32
33
counts
45.0
46.5
2θ (deg)